

Weak Learning Failure Alone Is Not Gradient Starvation: Matched Counterfactuals and Exact Finite-Width Dynamics

Anonymous authors

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Abstract

A weak predictive feature can remain poorly learned because its isolated learning is inherently slow, a stronger cue suppresses cross-entropy sample weights, or coupled parameter geometry actively penalizes its update. These explanations are not causally equivalent. We introduce a matched weak-only counterfactual: the paired models share initialization, data sequence, weak coordinates, and noise; only the strong cue is ablated. Causal starvation requires both outcome suppression and a nondegenerate weak-only first hit; gate failure leaves causal status indeterminate. For full-batch logistic gradient flow, we derive exact finite-width dynamics, $\dot{M}_w = c_w^\top q(r)/n$ and $\dot{r} = \Theta q(r)/n$, and the pathwise identity $\Delta = P - N$. In a frozen matched CIFAR-10 study, all 8/8 fresh seed blocks pass the causal certificate; the cue-trained model loses 19.9 percentage points on neutral inputs and 31.9 points under cue conflict. A common fresh linear head recovers part, but not all, of the neutral gap. Controlled recurrent studies further show that starvation is regime-dependent. Finally, Counterfactual Drift Correction (CDC) provides a unique minimum-norm instantaneous intervention in a fixed parameter chart, but operates as a local mechanism control rather than a globally superior optimizer.

1 Introduction

When a deep neural network learns one dominant feature while neglecting another predictive cue, the neglected signal is frequently described as “starved” [Pezeshki et al., 2021]. However, observing a small response to a weak feature during training is not by itself causal evidence of gradient starvation: the weak feature may simply be unlearnable under the chosen architecture, optimization horizon, or signal-to-noise ratio. The distinction is acute in recurrent neural networks, where cue strength, temporal separation, parameter sharing, and non-static geometry jointly govern representation learning [Bengio et al., 1994, Hochreiter and Schmidhuber, 1997, Cho et al., 2014].

To isolate the causal effect of a strong cue, we introduce a matched counterfactual intervention grounded in formal causal inference [Pearl, 2009, Peters et al., 2017]. A both-feature model $\mathcal{M}_B$ is paired with a weak-only model $\mathcal{M}_W$ that shares identical weight initialization, sample sequence, weak inputs, and optimizer state; only the strong input channel is zeroed out. Tracking the paired weak-feature response gap $\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau)$ and its time derivative $d(\tau) = \dot{\Delta}(\tau)$, positive d indicates *relative transfer*, negative d indicates *rate suppression*, and negative Δ indicates *outcome suppression*. We reserve the term *causal starvation* strictly for settings where outcome suppression occurs alongside a successful weak-only first hit of a prespecified target β within horizon H . When the weak-only gate fails, causal status is indeterminate: neither starvation nor its absence is established.

Four core contributions.

1. **Matched causal diagnosis.** We formalize the matched weak-only counterfactual and establish a two-condition causal certificate, proving that weak learning failure alone is insufficient to diagnose starvation.
2. **Exact finite-width dynamics.** We derive exact response-logit flow equations for full-batch logistic gradient flow, $\dot{M}_w = \frac{1}{n} c_w^\top q(r)$ and $\dot{r} = \frac{1}{n} \Theta q(r)$, separating cross-entropy weighting from parameter geometry. The pathwise identity $\Delta(t) = P(t) - N(t)$ establishes that rate suppression yields outcome suppression if and only if accumulated suppression exceeds accumulated transfer.

3. **Regime-dependent empirical evidence.** A frozen one-architecture CIFAR-10 confirmation yields causal certificates in 8/8 fresh paired seeds, persistent core-response deficits, and large neutral/conflict accuracy harm. In the audited recurrent results, tanh passes the at-hit certificate in 8/8 seeds (3/8 transfer then suppress; 5/8 suppress throughout), whereas GRU remains indeterminate because its weak-only gate fails in 8/8 seeds.
4. **Prediction limits and local mechanism control.** We characterize the scope of frozen-tangent models and prove that Counterfactual Drift Correction (CDC) is the unique minimum-norm instantaneous target-attaining intervention in a fixed parameter chart, providing localized mechanism control without asserting global optimizer superiority.

2 Related work

Feature starvation and shortcut learning. Cross-entropy loss can favor rapidly learnable features strongly enough to suppress gradient signal to slower predictive features, an effect termed gradient starvation [Pezeshki et al., 2021]. Shortcut learning, simplicity bias, and early memorization describe the broader empirical tendency of deep networks to exploit low-complexity, unintended cues at the expense of robust core features [Arpit et al., 2017, Geirhos et al., 2020, Shah et al., 2020, Hermann and Lampinen, 2020, Scimeca et al., 2022]. Invariant risk minimization and failure-guided debiasing seek representations invariant to spurious correlations [Arjovsky et al., 2019, Nam et al., 2020]. Our contribution is diagnostic: the matched weak-only counterfactual explicitly tests whether the neglected feature would have been learned in the absence of the strong cue, while direct drift and pathwise attribution distinguish transient rate suppression from adverse outcome suppression.

Methods such as Group DRO, Just Train Twice, and last-layer retraining target worst-group performance under spurious correlations [Sagawa et al., 2020, Liu et al., 2021b, Kirichenko et al., 2023]. We address a distinct question—causal attribution of feature-level learning dynamics—and do not propose a competing robustness benchmark. Our semi-real study evaluates controlled cue ablation on standard image benchmarks [LeCun et al., 1998, Xiao et al., 2017], preserving real core pixels while isolating causal feature interactions.

Implicit bias and margin dynamics. On separable classification problems trained with exponential or logistic losses, gradient flow exhibits an implicit bias toward directional maximum-margin separators in linear and homogeneous models [Soudry et al., 2018, Gunasekar et al., 2018, Ji and Telgarsky, 2019]. While asymptotic margin maximization governs directional convergence at infinite time, gradient starvation is an optimization-time transient phenomenon where rapid growth of dominant logit responses diminishes sample weights before secondary features can develop. Our exact finite-width equations capture this trajectory-level competition directly.

Wide and recurrent learning dynamics. Neural tangent kernels (NTK) characterize infinite-width gradient flow in the lazy training regime [Jacot et al., 2018, Du et al., 2019, Arora et al., 2019, Chizat et al., 2019], and wide-network linearization motivates frozen-tangent approximations [Lee et al., 2019]. Beyond the lazy regime, feature learning limits describe representation movement in overparameterized models [Yang and Hu, 2021, Bordelon and Pehlevan, 2022]. Our response–logit equations are instead exact at finite width for arbitrary trajectories. Freezing their kernels defines a nonlinear logistic tangent surrogate, but transferring its predictions to a trained network requires kernel-movement bounds that we do not establish for tanh RNNs or GRUs. Low-dimensional overlap theories for low-rank RNNs demonstrate that loss-invisible modes retain training history [Ger and Barak, 2026], echoing classical insights on disordered recurrent dynamics [Sompolinsky et al., 1988] and cautioning against closure in only the visible response coordinates. Mean-field descriptions of structure and disorder in trained recurrent circuits [Clark et al., 2026] study a distinct long-time equilibrium setting. We therefore do not claim a general recurrent optimization-time mean-field theory.

Gradient surgery and multi-task optimization. Multi-task learning often suffers from destructive gradient interference, inspiring multi-objective formulations including MGDA [Sener and Koltun, 2018], conflict-averse descent (CAGrad) [Liu et al., 2021a], gradient sign dropout (GradDrop) [Chen et al., 2020],

Table 1: Mathematical terminology and causal interpretation in the paired diagnostic framework.

Quantity	Sign	Causal / Dynamical Interpretation
$d(\tau) = \dot{\Delta}(\tau)$	> 0	Relative transfer: strong cue accelerates instantaneous weak learning.
$d(\tau) = \dot{\Delta}(\tau)$	< 0	Rate suppression: strong cue decelerates instantaneous weak learning.
$\Delta(\tau)$	< 0	Outcome suppression: accumulated weak response is worse under both cues.
$\tau_W^*(\beta)$	$\in (0, H]$	Weak-only learnable: weak-only control achieves target β within horizon H .
$\tau_W^*(\beta)$	$> H / \infty$	Weak-only unlearnable: gate fails; causal status is <i>indeterminate</i> .

rotational alignment (Rotograd) [Javaloy and Valera, 2022], and bargaining game approaches [Navon et al., 2022]. PCGrad projects conflicting task gradients onto orthogonal subspaces [Yu et al., 2020]; Bloop stabilizes auxiliary projections using an exponential moving average [Hsieh et al., 2024]. CDC also employs a projection, but its protected direction is the gradient of a strong-feature response and its target is derived from the matched weak-only trajectory. That extra counterfactual requires a constructible ablation, a second model, lockstep training, and full-batch gradients. Our comparisons use adapted shadow-target variants and support neither equivalence to the original methods nor broad CDC superiority.

3 Matched causal diagnosis

3.1 The matched weak-only counterfactual

Let condition $C \in \{B, W\}$ denote the both-feature and weak-only training runs, respectively. The two models share an identical parameter initialization $\theta_B(0) = \theta_W(0) = \theta_0 \in \mathbb{R}^P$, identical training sequence samples $(x_i, y_i)_{i=1}^n$, and identical optimization hyperparameters. The only difference is an intervention on the input: in W , the strong cue channel is set identically to zero ($x_{s,i} \equiv 0$).

Let $M_s^C(\theta), M_w^C(\theta)$ be differentiable scalar functionals measuring the model’s response to isolated strong and weak cues. For linear models, these correspond to channel-localized mode projections; for nonlinear recurrent models, they are defined via odd symmetric unit-probe contrasts: $M_a^C(\theta) = [f_C(x_a^+; \theta) - f_C(x_a^-; \theta)]/2$. For logistic gradient flow on full-batch signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$, the empirical loss is:

$$L_C(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-r_i^C(\theta)} \right), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (1)$$

The paired response gap $\Delta(\tau)$ and instantaneous equal-time relative drift $d(\tau)$ are:

$$\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau), \quad d(\tau) = \dot{\Delta}(\tau) = \dot{M}_w^B(\tau) - \dot{M}_w^W(\tau). \quad (2)$$

Since $\theta_B(0) = \theta_W(0)$, the gap begins at zero: $\Delta(0) = 0$.

3.2 Transfer, rate suppression, and outcome suppression

Table 1 summarizes the formal interpretation of signs in the paired framework.

3.3 Formal causal-starvation certificate

We now state the central definition of the paper.

Definition 1 (Causal starvation certificate). Fix a target response threshold $\beta > 0$ and optimization horizon $H > 0$. Let $\tau_W^*(\beta) = \inf\{\tau \in [0, H] : M_w^W(\tau) \geq \beta\}$ denote the first hitting time of the weak-only control, with $\inf \emptyset = \infty$. A training run exhibits a **causal starvation certificate** if and only if:

1. **Nondegenerate weak-only learnability:** $\tau_W^*(\beta) \in (0, H]$ with $\beta > M_w^W(0)$; and
2. **Outcome suppression at target:** $\Delta(\tau_W^*(\beta)) < 0$.

If $\tau_W^*(\beta) > H$, the weak-only gate fails and the causal status is **indeterminate**. If $\beta \leq M_w^W(0)$, the target is met at initialization (degenerate) and licenses no causal claim.

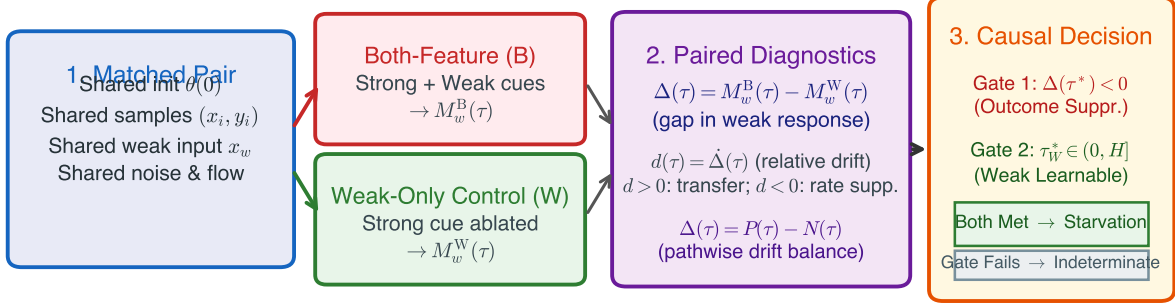


Figure 1: Schematic of the matched counterfactual causal diagnosis. A both-feature model $\mathcal{M}_B$ and weak-only control $\mathcal{M}_W$ share initialization $\theta(0)$, data samples (x_i, y_i) , and noise. The strong cue is ablated in W. Causal starvation is certified if and only if both outcome suppression ($\Delta < 0$) and the weak-only learnability gate ($\tau_W^* \leq H$) are satisfied.

4 Exact dynamics and pathwise attribution

4.1 Exact finite-width response–logit dynamics

We establish that the coupling between sample weighting and parameter geometry is an exact finite-width chain-rule identity rather than an asymptotic mean-field closure.

Proposition 1 (Exact finite-width response–logit dynamics). *Under full-batch logistic gradient flow (Eq. 1) at any finite width and parameter dimension P , let $J^C(\tau) \in \mathbb{R}^{n \times P}$ have rows $\nabla r_i^C(\theta_C(\tau))^\top$. Define the empirical tangent kernel $\Theta^C(\tau) = J^C(\tau)J^C(\tau)^\top \in \mathbb{R}^{n \times n}$, response-coupling vector $c_w^C(\tau) = J^C(\tau)\nabla M_w^C(\theta_C(\tau)) \in \mathbb{R}^n$, and sigmoid sample-weight vector $q(r^C)_i = \sigma(-r_i^C)$. Then:*

$$\dot{M}_w^C(\tau) = \frac{1}{n} c_w^C(\tau)^\top q(r^C(\tau)), \quad \dot{r}^C(\tau) = \frac{1}{n} \Theta^C(\tau) q(r^C(\tau)). \quad (3)$$

Proof Sketch. By chain rule, $\dot{\theta}_C = -\nabla L_C = \frac{1}{n} \sum_i \sigma(-r_i^C) \nabla r_i^C = \frac{1}{n} (J^C)^\top q(r^C)$. Differentiating M_w^C and r_i^C along $\dot{\theta}_C$ immediately gives Eq. (3). Full proof in Appendix C.1. $\square$

Interpretation. Eq. (3) factorizes learning velocity into two distinct terms:

- $q(r)$: The cross-entropy weighting vector. As logits r_i grow, $q_i \rightarrow 0$, diminishing the effective learning rate.
- c_w : The response geometry vector, reflecting alignment between gradients of M_w and sample logits r_i .

Corollary 1 (Closed projected flow under exact realization). *If signed sample logits admit the exact parameter-independent decomposition $r_i^C = \sum_b z_{ib}^C M_b^C + r_i^\perp$, then:*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C + R_a^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n} \sum_i z_{ib}^C \sigma(-r_i^C), \quad (4)$$

where $R_a^C = \frac{1}{n} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle$. When realization is exact ($r^\perp \equiv 0$), $R_a^C \equiv 0$ and the response dynamics close autonomously.

4.2 Cumulative drift balance

Proposition 2 (Cumulative drift balance). *Let $\Delta(\tau)$ be absolutely continuous on $[0, H]$ with $\Delta(0) = 0$ and $d(\tau) = \dot{\Delta}(\tau)$ almost everywhere. Define positive and negative drift components:*

$$P(t) = \int_0^t [d(u)]_+ du, \quad N(t) = \int_0^t [-d(u)]_+ du. \quad (5)$$

Then for all $t \in [0, H]$:

$$\Delta(t) = P(t) - N(t), \quad \text{and} \quad \Delta(t) < 0 \iff N(t) > P(t). \quad (6)$$

Proof. Since $\Delta(t) = \Delta(0) + \int_0^t d(u) du = \int_0^t ([d(u)]_+ - [-d(u)]_+) du = P(t) - N(t)$. Full details in Appendix C.2. $\square$

Core implication. A negative relative rate ($d(t) < 0$) does *not* imply outcome suppression ($\Delta(t) < 0$) unless the accumulated negative drift $N(t)$ repays all accumulated positive transfer $P(t)$.

Corollary 2 (Single-crossing tail-area criterion). *Suppose there exists $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ on (τ_d, H) . Let $A_+ = P(\tau_d) = \Delta(\tau_d) > 0$. Then for $t \geq \tau_d$, $\Delta(t) = A_+ - N(t)$. Strict outcome suppression occurs by horizon H if and only if $N(H) > A_+$.*

4.3 Scope of tangent prediction and movement bounds

Freezing Θ and c_w at initialization defines a nonlinear logistic tangent model: $\dot{r} = \frac{1}{n} \Theta_0 q(\bar{r})$, $\dot{M}_w = \frac{1}{n} c_{w,0}^\top q(\bar{r})$.

Theorem 1 (Conditional tangent comparison). *Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c_w(u) - c_{w,0}\| \leq B_c$, and $\|q(r)\| \leq V$. Let $a = \|\Theta_0\|/(4n)$, $\phi_a(t) = (e^{at} - 1)/a$, and $\psi_a(t) = (e^{at} - 1 - at)/a^2$. Then:*

$$|M_w(t) - \bar{M}_w(t)| \leq \frac{B_c V}{n} t + \frac{\|c_{w,0}\| B_\Theta V}{4n^2} \psi_a(t). \quad (7)$$

Theorem 1 is strictly conditional: establishing non-vacuous bounds B_Θ, B_c for nonlinear recurrent networks (tanh/GRU) remains an open challenge. Thus, tangent predictions operate as empirical event heuristics rather than calibrated trajectory guarantees.

5 Prediction and Counterfactual Drift Correction

5.1 Realized-initialization rate-gap criterion

For the noiseless rank-one dense linear RNN ($h_{k+1} = Wh_k + Bx_k$, $f(x) = c^\top h_T$), we compute the exact equal-time initial gap.

Theorem 2 (Realized-initialization rate-gap criterion). *Let the strong cue have strength $\rho > 0$ at lag 0, and the weak cue have lag $\ell \geq 0$. Define $K = G_{ws}(0) = b_s^\top W^\ell b_w$, $A = G_{ww}(0) = \|W^\ell b_w\|^2 + \|(W^\ell)^\top c\|^2 + \|S_w\|_F^2$, $s_B = \sigma(-[\rho M_s(0) + M_w(0)])$, and $s_W = \sigma(-M_w(0))$. The initial rate gap is:*

$$d_0 = s_B(\rho K + A) - s_W A, \quad d_0 > 0 \iff K > \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right). \quad (8)$$

Since $\Delta(0) = 0$, $d_0 < 0 \implies \Delta(t) < 0$ for sufficiently small $t > 0$.

This criterion is exact at $t = 0$ but local: it does not constitute a positive-lag global phase predictor.

5.2 Counterfactual Drift Correction (CDC)

To evaluate whether the diagnosed mechanism can be controlled locally, we introduce Counterfactual Drift Correction (CDC). In a fixed Euclidean parameter chart, let $a = \nabla m_s(\theta)$, $p = \nabla m_w(\theta)$, and $v = -\nabla L(\theta)$. Project p onto the orthogonal complement of a :

$$q = p - \frac{p \cdot a}{\|a\|^2} a \quad (a \neq 0), \quad \delta = [F_w^W - p \cdot v]_+, \quad \alpha = \frac{\delta}{\|q\|^2} \quad (q \neq 0). \quad (9)$$

The corrected parameter update is $v' = v + \alpha q$.

Theorem 3 (CDC local guarantee). *In the fixed parameter chart:*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly (strong drift is unaffected).
2. **Minimum Euclidean norm:** $d^* = \alpha q$ is the unique minimum-norm correction attaining the weak-drift lower bound $p \cdot (v + d) \geq F_w^W$ subject to $a \cdot d = 0$.
3. **Local finite-step deviation:** For L -Lipschitz ∇m_s , $|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2) = O(\eta^2)$.

CDC provides first-order instantaneous mechanism control; it is not a claim of global optimization superiority.

6 Experimental setup

6.1 Controlled linear study

We evaluate linear RNNs (width 96, spectral gain $g = 0.5$, full-batch gradient descent) across a grid of strength ratios $\rho \in \{1, 2, 4, 8\}$ and temporal lags $\ell \in \{0, 2, 4, 8\}$, with 8 paired seeds per cell (seed IDs 4–11). The learnability gate is set to $\beta = 0.25$ with horizon $H = 5.0$. A sign-reversal intervention reverses the strong cue at $(\rho, \text{lag}) = (4, 4)$; it is an equivariance check, not a no-cue negative control.

6.2 Nonlinear recurrent study

We evaluate tanh RNNs and GRUs at width 32, gain $g = 0.8$, cue noise $\sigma_{\text{noise}} = 0.1$, $(\rho, \text{lag}) = (4, 2)$, and target $\beta = 0.5$ over 8 paired seeds. We evaluate instantaneous drift via direct automatic differentiation ($F_w = \nabla M_w \cdot \hat{\theta}$).

6.3 Frozen matched CIFAR-10 confirmation

Before inspecting the cue-present outcomes, we froze one CNN4-GN architecture, eight fresh paired seeds, SGD with momentum, a 4,710-update cosine schedule, and all endpoints. B appends a removable label-correlated ribbon (reliability 0.9, amplitude 0.5) outside the 32×32 core; W receives the identical core and a neutral zero-valued ribbon after preprocessing. Within each seed block, B and W share initialization, source-image split, minibatch order, augmentation randomness, and optimizer configuration. The primary response is balanced Brier skill on a fixed neutral diagnostic set, with $\beta = 0.5$ and scale $S = 0.75$. Final behavior uses neutral, random, consistent, and nine within-image conflict cues. E4-lite freezes each final encoder and fits the same balanced linear-head protocol on common neutral inputs. This discrete minibatch experiment tests the matched causal certificate, not the continuous-time tail-area identity.

6.4 Tangent and intervention evaluations

- **Held-out tangent event test:** 64 crossed records (4 data seeds $\times$ 8 model seeds) evaluating drift and response crossing classifications.
- **CDC mechanism control:** Comparison of ERM, Spectral Decoupling, CDC, and adapted shadow-target variants (Bloop-style, PCGrad-style) across 256 method runs.

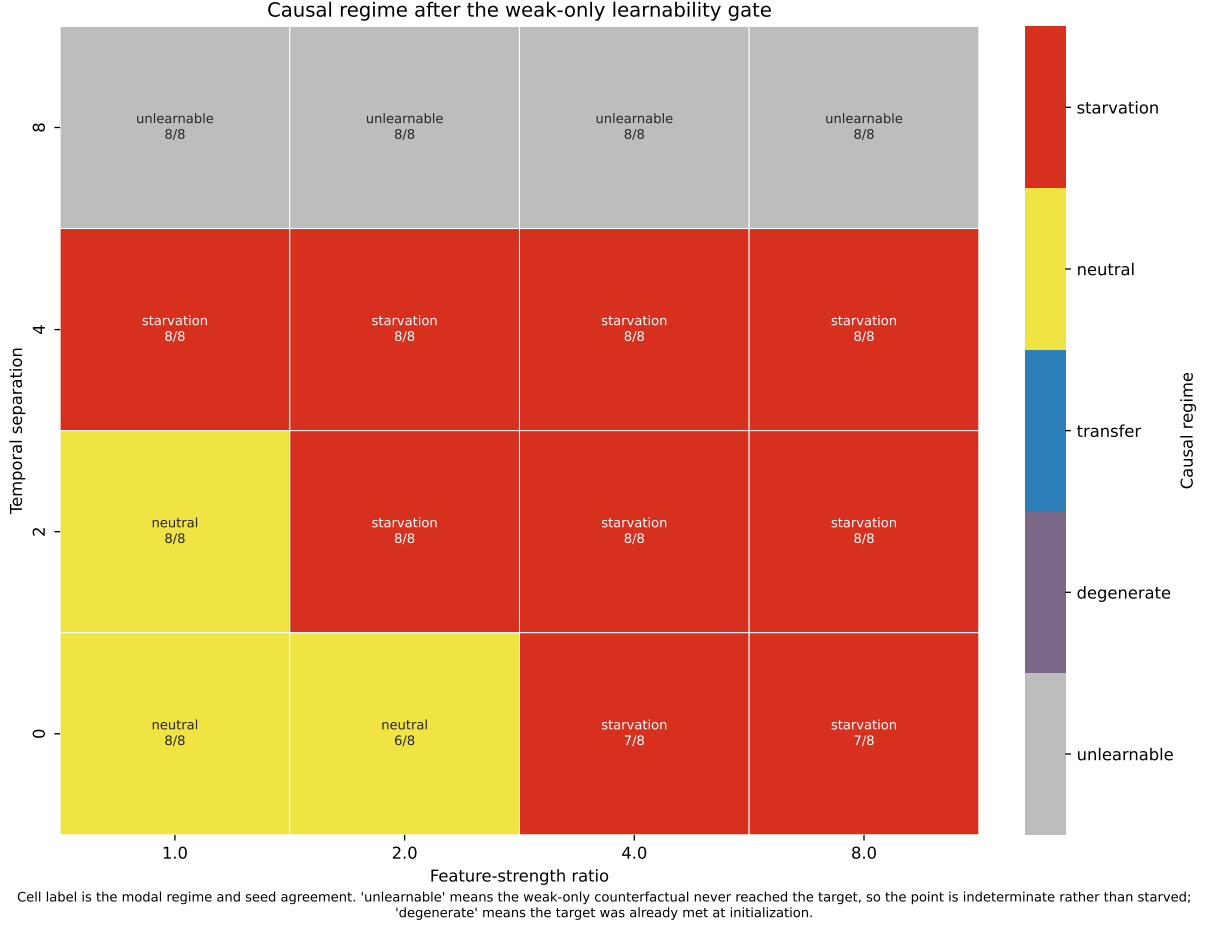


Figure 2: Controlled dense-linear weak-AUC deficits and gate status across $\rho \in \{1, 2, 4, 8\}$ and lag. The plotted historical regime labels are threshold-delay categories, not the at-hit certificate. The audited at-hit counts are 31/32, 32/32, 32/32, and 0/32 for lags 0, 2, 4, and 8.

- **Semi-real image study:** Controlled cue intervention on MNIST (3-vs-8) and FashionMNIST (0-vs-6) over 64 crossed records.

7 Results

7.1 Regime-dependent causal diagnosis

Figure 2 and Table 2 present the audited controlled linear grid. At lags 0, 2, and 4, all 32 cells have positive paired weak-AUC deficits and pass the weak-only gate. The stricter at-hit certificate passes in 31/32 lag-0 cells and 32/32 cells at lags 2 and 4; the lag-0 exception later becomes negative and passes the broader any-time endpoint. At lag 8, all 32 cells fail the learnability gate and are causally indeterminate, even though 27/32 have positive signed AUC. The mean deficit peaks at lag 2 (8.93) rather than decaying monotonically.

7.2 Nonlinear crossings versus causal certificates

Figure 3 illustrates dynamic trajectories at the fixed nonlinear cell $(\rho, \text{lag}) = (4, 2)$.

- **Tanh RNN:** Passes the weak-only gate and at-hit certificate in 8/8 seeds: 3/8 transfer before suppressing, while 5/8 suppress throughout without a crossing.

Table 2: Linear grid paired weak-AUC deficits $\mathcal{A}_w$, weak-only hitting times τ_W^* , and causal status.

Lag	Strength ρ	Mean $\mathcal{A}_w$ (95% CI)	Weak Gate ($\tau_W^* \leq 5$)	Causal Certificate
0	1–8	5.76 [5.01, 6.51]	32/32 Passed	31/32 Certified
2	1–8	8.93 [8.29, 9.57]	32/32 Passed	32/32 Certified
4	1–8	6.11 [5.31, 6.91]	32/32 Passed	32/32 Certified
8	1–8	0.0015 [0.0010, 0.0020]	0/32 Passed	0/32 (Indeterminate)
Neg. Control	4.0 (Lag 4)	−4.04 [−4.50, −3.58]	8/8 Passed	0/8 (Transfer)

Table 3: Held-out evaluation of the frozen-tangent surrogate (64 crossed records).

Evaluation Target	Endpoint	Accuracy	95% Bootstrap CI
Held-out Event Test	Drift crossing ($d(\tau) < 0$)	60/64 (93.8%)	[87.5%, 98.4%]
Held-out Event Test	Response crossing ($\Delta(\tau) < 0$)	64/64 (100.0%)	[100.0%, 100.0%]
Held-out Causal Gate	Causal certificate	0/64 (0.0%)	[0.0%, 0.0%]

- **GRU:** Exhibits direct-autograd drift crossings in 5/8 seeds and response crossings in 5/8 seeds. However, the weak-only model never reaches $\beta = 0.5$ ($\tau_W^* > 5.0$), resulting in **0/8 causal certificates**.

This establishes empirically that crossing subtype and causal certificate are distinct: a run can be starved without crossing, while a crossing with a failed weak-only gate remains indeterminate.

7.3 A removable image cue causally suppresses core learning

All eight frozen CIFAR-10 pairs pass the nondegenerate W gate and the primary at-hit certificate. Mean normalized signed deficit AUC is 0.489 (95% CI [0.413, 0.564]), compared with the frozen target 0.10, and the material deficit persists for 98.3% of training ([97.6%, 99.0%]). At the final checkpoint, B trails W by 19.9 percentage points on neutral inputs ([−22.6, −17.2]), 27.6 points under random cues ([−28.7, −26.4]), and 31.9 points under cue conflict ([−33.0, −30.8]), while gaining 16.8 points when the cue is label-consistent ([15.8, 17.9]).

With the original heads, the neutral B–W gap is −19.9 points. Equal fresh balanced linear heads reduce it to −10.8 points ([−11.1, −10.4]), a 9.1-point recovery ([6.5, 11.8]). Thus, part of the harm is readout underuse, while a substantial remainder is reduced linear accessibility under this probe; this is not evidence of information destruction.

7.4 Frozen tangent predicts crossing events, not causal status

Table 3 reports the held-out performance of the frozen-tangent surrogate. The surrogate accurately classifies binary drift crossings (60/64, 93.8%) and response crossings (64/64, 100.0%). However, because weak-only learnability cannot be verified without trajectory integration, the tangent surrogate cannot predict causal starvation certificates.

7.5 Local mechanism control with CDC

Table E.2 summarizes the dense-linear mechanism control comparison across 8 paired seeds. All shadow-based interventions eliminate the weak AUC gap from ERM’s 10.52 to ≈ 0.00 . However, CDC uniquely preserves instantaneous strong drift (perturbation 4.77×10^{-7} , compared to $1.48 \times 10^{-1} - 3.75 \times 10^{-1}$ for alternatives). As predicted by theory, this local one-step guarantee does not imply maximum final strong response (1.042 vs. 1.295 for unconstrained rescue).

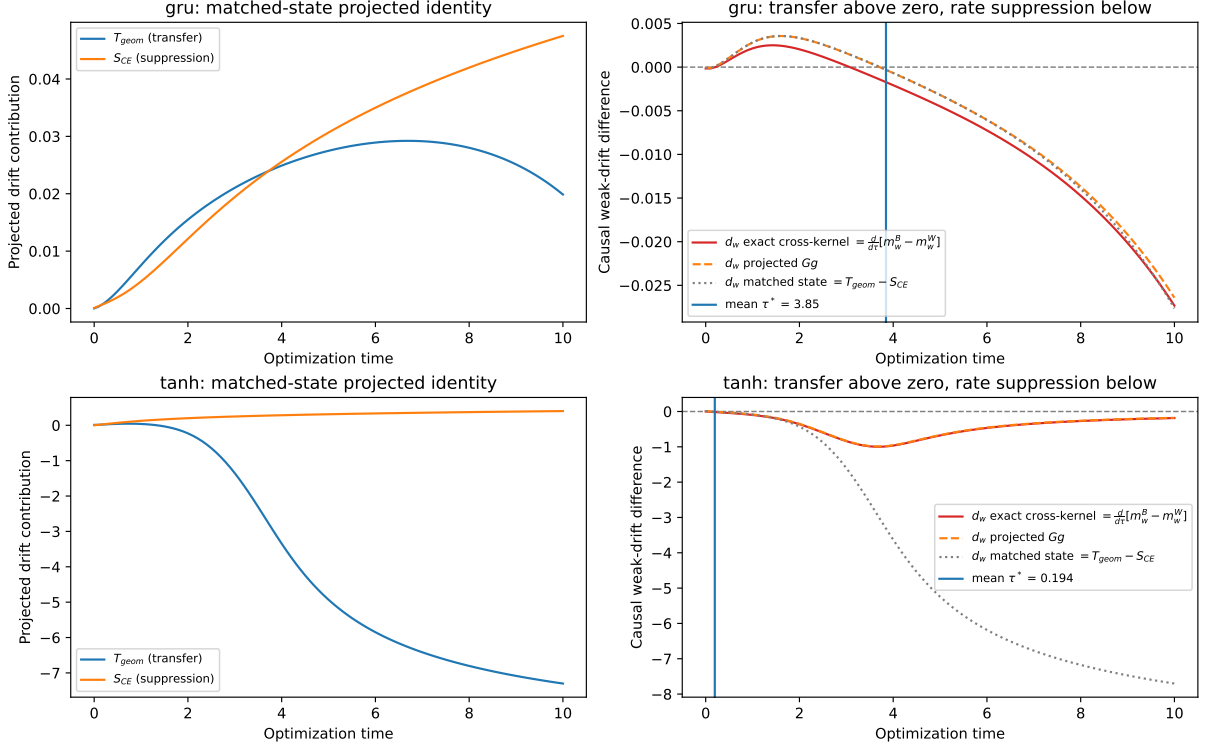


Figure 3: Paired nonlinear trajectories at $(\rho, \text{lag}) = (4, 2)$. Tanh (left) exhibits a certified causal starvation event. GRU (right) exhibits transient rate and response crossings but fails the weak-only gate, remaining causally indeterminate.

8 Limitations and conclusion

8.1 Scope and limitations

The empirical evidence is restricted to controlled synthetic grids and one preregistered CIFAR-10 intervention using an artificial ribbon cue and one architecture. It establishes a controlled existence result and diagnostic utility, not the prevalence of starvation under natural shortcuts or a universal architecture ranking. The CIFAR experiment uses minibatch SGD and therefore does not verify the continuous-time tail-area theorem. The tangent surrogate requires unproved movement bounds to supply calibrated trajectories, and CDC requires an exact matched shadow model under full-batch gradient flow in a fixed parameter chart.

8.2 Conclusion

Weak learning failure alone is not gradient starvation. By introducing a matched weak-only counterfactual, deriving exact finite-width dynamics $\dot{M}_w = \frac{1}{n} c_w^\top q(r)$, and establishing cumulative drift balance $\Delta = P - N$, we establish a rigorous causal and dynamical framework for feature starvation. The resulting causal chain—exact dynamics $\rightarrow$ matched intervention $\rightarrow$ pathwise attribution $\rightarrow$ local mechanism control—disciplines the study of feature competition in deep networks.

Reproducibility statement

Complete source code, configurations, environment specifications, and content-addressed result manifests for the linear, recurrent, tangent, CDC, and frozen CIFAR-10 experiments are provided in the supplementary repository and documented in Appendix D.

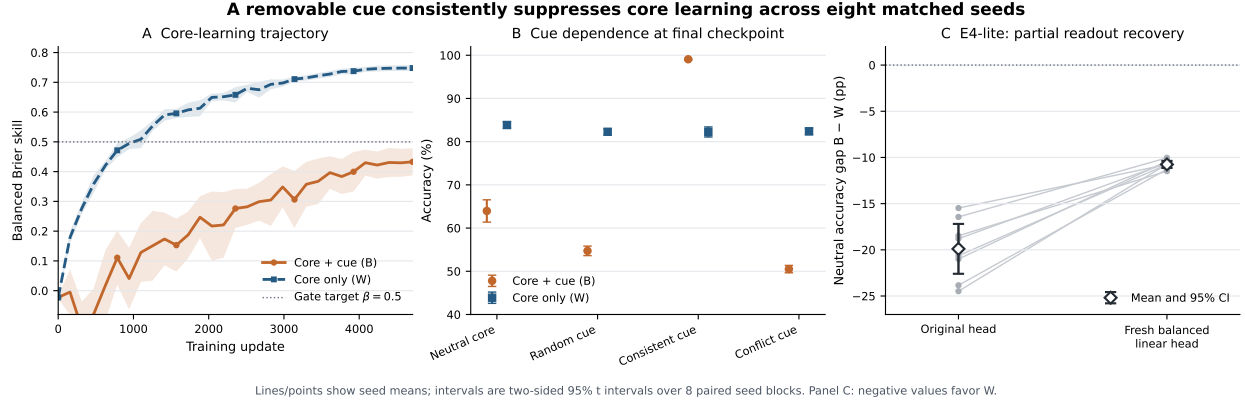


Figure 4: Frozen matched CIFAR-10 confirmation across eight paired seed blocks. Lines and points are seed means with two-sided 95% t intervals. B contains the removable cue; W contains the identical core with a neutral cue channel. Every seed passes the W gate and at-hit certificate. The fresh-head test recovers part, but not all, of the neutral gap.

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AI use statement

Generative AI tools (OpenAI Codex/ChatGPT, Google Gemini, and Anthropic Claude) assisted with code refactoring, experiment infrastructure, mathematical formatting verification, and manuscript drafting and editing. The authors reviewed all AI-assisted material, independently checked the reported artifacts and mathematical proofs, and take full responsibility for the contents of the paper.

A Full definitions and notation

A.1 Paired logistic gradient flow

Let $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ be a dataset with binary labels $y_i \in \{-1, +1\}$ and inputs $x_i = (x_{s,i}, x_{w,i}, x_{bg,i}) \in \mathbb{R}^{d_s + d_w + d_{bg}}$. For condition $C \in \{B, W\}$, the parameter $\theta_C(\tau) \in \mathbb{R}^P$ evolves under full-batch gradient flow:

$$L_C(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-y_i f_C(x_i; \theta)} \right), \quad \frac{d\theta_C}{d\tau} = -\nabla L_C(\theta_C), \quad (\text{A.1})$$

where $f_B(x; \theta) = f(x; \theta)$ and $f_W(x; \theta) = f((0, x_w, x_{bg}); \theta)$. Both systems share initialization $\theta_B(0) = \theta_W(0) = \theta_0$.

A.2 Hitting time metrics and weak-AUC deficit

For any response functional $M(\tau)$ and target threshold $\beta > 0$:

$$\tau(\beta; H) = \inf\{\tau \in [0, H] : M(\tau) \geq \beta\}, \quad (\inf \emptyset = \infty). \quad (\text{A.2})$$

The paired weak-AUC deficit over horizon H is defined as:

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) d\tau = - \int_0^H \Delta(\tau) d\tau. \quad (\text{A.3})$$

B Additional theoretical results

B.1 Closed projected flow under exact realization

Proposition B.3 (Closed projected flow). *If signed sample logits satisfy $r_i^C(\theta) = \sum_b z_{ib}^C M_b^C(\theta) + r_i^\perp(\theta)$, then:*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C + R_a^C, \quad (\text{B.1})$$

where $G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle$, $g_b^C = \frac{1}{n} \sum_i z_{ib}^C \sigma(-r_i^C)$, and $R_a^C = \frac{1}{n} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle$. When realization is exact ($r_i^\perp \equiv 0$), $R_a^C \equiv 0$ and the response dynamics close autonomously.

B.2 Log drift-ratio identity

Theorem B.1 (Log drift-ratio identity). *For rank-one positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, write $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$ where $H_B = \rho G_{ws}^B + G_{ww}^B$ and $H_W = G_{ww}^W$. Define $\Psi = \log(F_w^B / F_w^W)$. Then $\text{sign}(d) = \text{sign}(\Psi)$, and:*

$$\dot{\Psi} = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \frac{d}{d\tau} \log \left(\frac{H_B}{H_W} \right), \quad (\text{B.2})$$

where $Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0$.

B.3 Every-width six-scalar solvable zero-lag limit

Theorem B.2 (Every-width six-scalar closure at zero lag). *At zero temporal lag and zero background noise in the dense linear RNN, the recurrent weight matrix W is loss-invisible ($\nabla_W L = 0$). With $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$, the dynamics close at every width N :*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad \dot{w} = 2 \sum_b g_b M_b. \quad (\text{B.3})$$

C Proofs

C.1 Proof of Proposition 1

Under full-batch logistic gradient flow, $\dot{\theta} = -\nabla L(\theta) = \frac{1}{n} \sum_{i=1}^n \sigma(-r_i) \nabla r_i(\theta) = \frac{1}{n} J(\theta)^\top q(r)$. By chain rule, for any response $M_w(\theta)$: $\dot{M}_w = \langle \nabla M_w, \dot{\theta} \rangle = \frac{1}{n} \langle \nabla M_w, J^\top q(r) \rangle = \frac{1}{n} (J \nabla M_w)^\top q(r) = \frac{1}{n} c_w^\top q(r)$. Similarly, $\dot{r}_i = \langle \nabla r_i, \dot{\theta} \rangle = \frac{1}{n} \sum_j \langle \nabla r_i, \nabla r_j \rangle q_j = \frac{1}{n} (\Theta q(r))_i$. $\square$

C.2 Proof of Proposition 2

Since $\Delta(0) = 0$ and $\Delta(\tau)$ is absolutely continuous, $\Delta(t) = \int_0^t \dot{\Delta}(u) du = \int_0^t d(u) du$. Writing $d(u) = [d(u)]_+ - [-d(u)]_+$ gives $\Delta(t) = \int_0^t [d(u)]_+ du - \int_0^t [-d(u)]_+ du = P(t) - N(t)$. Thus $\Delta(t) < 0 \iff P(t) - N(t) < 0 \iff N(t) > P(t)$. $\square$

C.3 Proof of Theorem 1

Let $e_r(t) = r(t) - \bar{r}(t)$ and $e_M(t) = M(t) - \bar{M}(t)$. Differentiating: $\dot{e}_r = \frac{1}{n} [\Theta(t)q(r(t)) - \Theta_0 q(\bar{r}(t))] = \frac{1}{n} [\Theta(t) - \Theta_0]q(r(t)) + \frac{1}{n} \Theta_0 [q(r(t)) - q(\bar{r}(t))]$. Since σ' is bounded by $1/4$, $\|q(r) - q(\bar{r})\| \leq \frac{1}{4} \|e_r\|$. Applying Grönwall's inequality with $a = \|\Theta_0\|/(4n)$ yields the bounds in Eqs. (7). $\square$

C.4 Proof of Theorem 2

At $\tau = 0$, $\theta_B(0) = \theta_W(0)$. Using the dense linear block gradients with $W(0), B(0), c(0), G_{ws}(0) = b_s^\top W^\ell b_w = K$ and $G_{ww}(0) = A$. Evaluating the initial weak drift in each condition: $\dot{M}_w^B(0) = s_B(\rho K + A)$ and $\dot{M}_w^W(0) = s_W A$. Subtracting gives $d_0 = s_B(\rho K + A) - s_W A$. Setting $d_0 > 0$ yields Eq. (8). $\square$

C.5 Proof of Theorem 3

For (1): $a \cdot v' = a \cdot (v + \alpha q) = a \cdot v + \alpha(a \cdot q) = a \cdot v$, since $a \cdot q = a \cdot (p - \frac{p \cdot a}{\|a\|^2} a) = 0$. For (2): The optimization problem is $\min_d \frac{1}{2} \|d\|^2$ subject to $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$. Forming the Lagrangian $L(d, \lambda_1, \lambda_2) = \frac{1}{2} \|d\|^2 - \lambda_1(a \cdot d) - \lambda_2(p \cdot d - \delta)$, stationarity gives $d^* = \lambda_1 a + \lambda_2 p$. Orthogonality $a \cdot d^* = 0$ forces $\lambda_1 = -\lambda_2 \frac{a \cdot p}{\|a\|^2}$, so $d^* = \lambda_2 q$. Enforcing $p \cdot d^* = \delta$ yields $\lambda_2 = \delta / \|q\|^2 = \alpha$, proving uniqueness. For (3): By Taylor expansion with Lipschitz gradient ∇m_s , $|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \eta |a \cdot (v' - v)| + \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2) = \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2)$. $\square$

D Experimental details and complete hyperparameters

Table D.1: Complete experimental hyperparameters across all study designs.

Hyperparameter	Linear Grid (§7.1)	Nonlinear Fixed-Cell (§7.2)	Tangent Held-Out (§7.3)	CDC Ablation (§7.4)
Architecture	Dense Linear RNN	Tanh RNN / GRU	Tanh RNN / GRU	Dense Linear RNN
Hidden width N	96	32	32	96
Recurrent Gain g	0.5	0.8	0.8	0.5
Input scaling σ_B	$1/\sqrt{N}$	$1/\sqrt{N}$	$1/\sqrt{N}$	$1/\sqrt{N}$
Sequence length T	10	10	10	10
Optimizer	Full-batch GD	Full-batch GD	Full-batch GD	Full-batch GD
Learning rate η	0.1	0.05	0.05	0.1
Horizon H	5.0 (50 steps)	5.0 (100 steps)	5.0 (100 steps)	5.0 (50 steps)
Target threshold β	0.25	0.5	0.5	0.25
Replication	8 seeds $\times$ 16 cells	8 seeds $\times$ 2 archs	4 data $\times$ 8 model seeds	8 paired seeds
Hardware	Apple M-series / CUDA	Apple M-series / CUDA	Apple M-series / CUDA	Apple M-series / CUDA

D.1 Frozen CIFAR-10 confirmation details

The confirmatory source commit is 0726aea12947a30eb60ae32fc9c1eb0a7a122660; the frozen configuration and DGX environment have SHA-256 digests 8f662f9982f8445b and 1c1e2d126093f449, respectively (prefixes shown). Runs use CIFAR-10 archive SHA-256 6d958be074577803. The split contains 40,000 training, 5,000 validation, 2,500 probe-fit, and 2,500 diagnostic-evaluation images. Full hashes, raw traces, and the independent audit are included in the supplementary artifacts.

Table D.2: Per-seed frozen CIFAR-10 outcomes. Duration is the longest material core-response deficit as a fraction of the 4,710-update horizon. Accuracy gaps are B–W on neutral inputs in percentage points.

Seed	W first hit	Signed deficit AUC	Duration	Original-head gap	Fresh-head gap
33001	942	0.351	0.986	−16.44	−10.04
33002	1256	0.573	0.990	−23.84	−11.28
33003	785	0.495	0.974	−18.80	−10.60
33004	1099	0.616	0.989	−24.48	−10.72
33005	942	0.442	0.968	−18.48	−11.48
33006	942	0.495	0.986	−21.00	−10.40
33007	1256	0.543	0.978	−20.64	−10.76
33008	1099	0.393	0.993	−15.48	−10.76

E Supplementary experiments and ablations

E.1 Width trends and finite-network closure validation

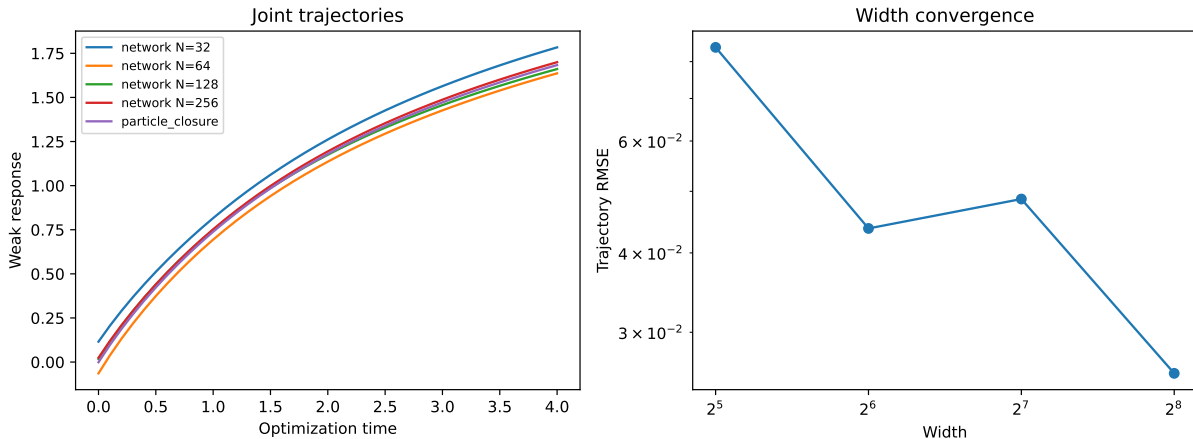


Figure E.1: Finite-width agreement with held-out empirical closure reference across widths $N \in \{32, 64, 128, 256\}$.

E.2 Semi-real generated-cue study

E.3 Threshold sensitivity and β -sweeps

E.4 Full 8-method mitigation ablation

Table E.1: Semi-real generated-cue study on MNIST (3-vs-8) and FashionMNIST (0-vs-6) across 64 crossed records.

Dataset	Weak-Only Learnable	Causal Certificate	Mean Weak-AUC Gap (95% CI)
MNIST 3-vs-8	0/32	0/32	−0.00124 [−0.00344, 0.00102]
FashionMNIST 0-vs-6	0/32	0/32	−0.00011 [−0.00413, 0.00318]

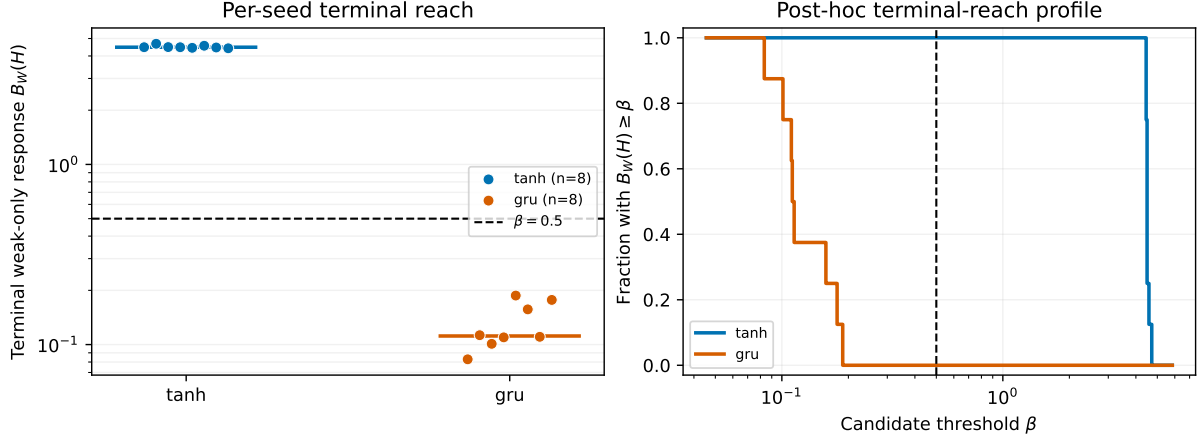


Figure E.2: Post-hoc weak-only terminal response distributions and threshold sensitivity profiles across seeds.

Table E.2: Dense CDC ablation across 8 paired seeds. Shadow methods access a matched weak-only control.

Method	Shadow	Weak AUC Gap	Strong Drift Perturbation	Final M_s
ERM	No	10.516	—	1.364
Spectral Decoupling	No	9.023	—	0.829
CDC (Ours)	Yes	−0.0005	4.77×10^{-7}	1.042
Unconstrained Rescue	Yes	−0.0007	1.48×10^{-1}	1.295
Adapted Bloop-style	Yes	0.0002	3.75×10^{-1}	0.370
Adapted PCGrad-style	Yes	−0.0007	1.48×10^{-1}	1.295

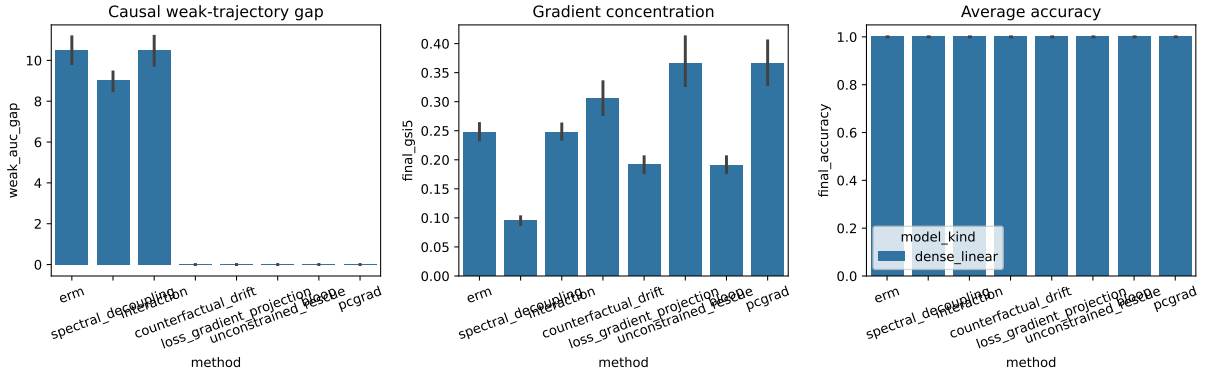


Figure E.3: Full eight-method dense-linear mitigation ablation comparing CDC with ERM, Spectral Decoupling, PCGrad, Bloop, and unconstrained rescue.